Insurance Analytics Project

Insurance Portfolio Risk & Profitability Analysis

Project Overview

Exploring a simulated insurance dataset

This project utilizes a dataset of **1,000,000 insurance policies** to analyze trends in policy sales, claims frequency, and overall profitability for strategic decision-making.

Business Context

Analyzing Insurance Sales and Claims

This project focuses on an insurance company’s analysis of policy sales, claims frequency, premium revenue versus liabilities, and loss ratio trends from 2024 to 2026.

Dataset Simulation

Policy Sales Details and Claims Rules

Policy Sales

The simulated dataset consists of 1,000,000 insurance policies, with vehicle values set at ₹100,000 and annual premiums of ₹100, ensuring a comprehensive analysis.

Tenure Distribution

Policies have varied tenure distributions: 20% for 1 year, 30% for 2 years, 40% for 3 years, and 10% for 4 years, impacting overall profitability.

Claims Rules

According to rules, 30% of policies will incur claims in 2025 based on specific purchase dates, with 10% of 4-year policies claimed early in 2026.

Project Objectives

Key Goals for Data Analysis

Total Premium

Calculate the overall **total premium** collected from policies to assess revenue generation and evaluate financial health over the specified periods.

Claims Analysis

Analyze claims frequency and trends to identify patterns and **evaluate the impact** on profitability, ensuring effective claims management and resource allocation.

Claim-to-Premium Ratio

Determine the **claim-to-premium ratio** to assess risk exposure, allowing the insurance company to adjust pricing strategies and enhance profitability.

Data Pipeline

The data pipeline illustrates seamless integration of data from Python to SQL and visualization in Power BI.

Python (Data Simulation)
↓
CSV Data Storage
↓
SQL Analysis
↓
Power BI Data Modeling
↓
Interactive Dashboard
↓
Business Insights

Table Structure

```sql
USE car_insurance;

CREATE TABLE policy_sales (

customer_id INT,
vehicle_id INT,
vehicle_value INT,
premium INT,
policy_purchase_date DATE,
policy_start_date DATE,
policy_end_date DATE,
policy_tenure INT

);

CREATE TABLE claims (

claim_id INT,
customer_id INT,
vehicle_id INT,
claim_amount INT,
claim_date DATE,
claim_type INT

);
```

Analytical Queries

Q 1. Calculate the total premium collected during the year 2024.

```sql
-- Q 1.  Calculate the total premium collected during the year 2024.

SELECT  SUM(premium) AS  total_premium
FROM policy_sales
WHERE YEAR(policy_purchase_date) = '2024'
```

100 % 2 0

Results | Messages

	total_premium
1	239834700

Q 2. Calculate the total claim cost for each year (2025 and 2026) with a monthly breakdown.

```sql
-- Q 2.  Calculate the total claim cost for each year (2025 and 2026) with a monthly breakdown.

SELECT
YEAR(claim_date) AS year,
MONTH(claim_date) AS month,
SUM(claim_amount) AS total_claim_cost
FROM claims
GROUP BY YEAR(claim_date), MONTH(claim_date)
ORDER BY year, month;
```

100 % 2 0

Results Messages

	year	month	total_claim_cost
1	2025	1	32730000
2	2025	2	32390000
3	2025	3	32310000
4	2025	4	32670000
5	2025	5	32830000
6	2025	6	33040000
7	2025	7	32200000
8	2025	8	32960000
9	2025	9	33050000
10	2025	10	32990000
11	2025	11	34040000
12	2025	12	32230000
13	2026	1	52250000
14	2026	2	47450000

Q 3. Calculate the claim cost to premium ratio for each policy tenure (1, 2, 3, and 4 years).

```sql
-- Q 3. Calculate the claim cost to premium ratio for each policy tenure (1, 2, 3, and 4 years).

SELECT
p.policy_tenure,
SUM(c.claim_amount) / SUM(p.premium) AS claim_ratio
FROM policy_sales p
LEFT JOIN claims c
ON p.vehicle_id = c.vehicle_id
GROUP BY p.policy_tenure;
```

100 % 2 0

Results Messages

	policy_tenure	claim_ratio
1	1	3
2	2	1
3	3	1
4	4	3

Q 4. Calculate the claim cost to premium ratio by the month in which the policy was sold (January-December 2024).

```sql
-- Q 4. Calculate the claim cost to premium ratio by the month in which the policy was sold  (January-December 2024).

SELECT
MONTH(policy_purchase_date) AS sale_month,
SUM(c.claim_amount) / SUM(p.premium) AS ratio
FROM policy_sales p
LEFT JOIN claims c
ON p.vehicle_id = c.vehicle_id
GROUP BY MONTH(policy_purchase_date);
```

Q 5. If every vehicle that has not yet made a claim eventually files exactly one claim during the remaining policy tenure, estimate the total potential claim liability.

```sql
GROUP BY MONTH(policy_purchase_date);

-- Q 5. If every vehicle that has not yet made a claim eventually files exactly one claim during the remaining policy tenure,

WITH total_vehicles AS (
    SELECT COUNT(*) AS total FROM policy_sales
),
claimed AS (
    SELECT COUNT(DISTINCT vehicle_id) AS claimed FROM claims
)
SELECT
    CAST((total - claimed) AS BIGINT) * 10000 AS future_claim_liability
FROM total_vehicles, claimed;
```

Q 6 Calculate the premium already earned by the company up to February 28, 2026.

```sql
-- Q 6  Calculate the premium already earned by the company up to February 28, 2026.

SELECT
SUM(
    (premium * 1.0 / DATEDIFF(day, policy_start_date, policy_end_date))
    *
    DATEDIFF(day, policy_start_date,
        CASE
            WHEN policy_end_date < '2026-02-28' THEN policy_end_date
            ELSE '2026-02-28'
        END
    )
) AS earned_premium
FROM policy_sales
WHERE policy_start_date <= '2026-02-28';


-- Part 4: Bonus Questions
```

% 2 0

Results | Messages

earned_premium
65902398.082017139467

Q 7. Identify which policy tenure appears most profitable.

```sql
SELECT
p.policy_tenure,
SUM(p.premium) AS total_premium,
SUM(c.claim_amount) AS claims,
SUM(p.premium) - COALESCE(SUM(c.claim_amount),0) AS profit
FROM policy_sales p
LEFT JOIN claims c
ON p.vehicle_id = c.vehicle_id
GROUP BY p.policy_tenure;
```

100 % 2 0

Results Messages

	policy_tenure	total_premium	claims	profit
1	1	20054900	79280000	-59225100
2	2	60051200	117630000	-57578800
3	3	119848200	157940000	-38091800
4	4	40043200	138290000	-98246800

Dashboard Insights

Findings

- Claim costs peak in January–February, then stabilize across the year.
- 2025 accounts for the majority of claim payouts (~₹393M).
- 3-year policies generate the highest claim exposure due to the largest customer base.
- 80% of claims are first-time claims, with only a small portion being repeat claims.
- Policy distribution is heavily concentrated in 3-year tenures (~40%).
- The loss ratio of 2.06 indicates claims exceed premium revenue, suggesting potential pricing adjustments may be required.

Business Impact

1. **Extremely High Loss Ratio**
   - The Loss Ratio = 2.06 (206%), meaning claims are more than double the premium collected.
   - Business Impact: The insurance portfolio is highly unprofitable, leading to financial losses.
2. **Claims Significantly Exceed Premiums**
   - Total Premium: 240M
   - Total Claims: 493M
   - Business Impact: The company is paying 253M more in claims than it earns, indicating pricing or risk assessment issues.
3. **Certain Policy Tenures Generate Higher Claims**
   - **P**olicy Tenure 3 and 4 show the highest claim values.
   - Business Impact: Long-tenure policies may represent higher risk customers or aging assets, increasing claim costs.
4. **High Claims Concentrated in Early Months**
   - January and February show very high claim amounts (~80–90M) compared to later months (~30M).
   - Business Impact: Possible seasonal risks or operational issues, such as weather, accidents, or claim surges.
5. **Majority of Claims Come from Claim Type 1**
   - Claim Type 1 contributes ~80% of claims.
   - Business Impact: This specific claim category is the primary driver of claim costs.

Tools and Technologies

Key Tools

- Python (Pandas, NumPy)
- SQL for Data Analysis and manage
- Power BI for Visualization
- BI Analytics for Insights

Thank You

I appreciate your attention